CSE 213: What to Study, in Order

Based on 11 past papers (CSE 213 2025–26, CSE 307 2011–24), weighted toward recent papers.

#	Topic	What to master	Slides	How often
TIER 1: MUST STUDY (about 40–55% of marks)				
1	**Behavioral patterns**	Observer, Strategy, Mediator, Command, State, Template, Chain of Resp. Compare State vs Strategy.	IJ Mam 2, 3	Every paper
2	**Creational patterns**	Factory, Abstract Factory, Builder, Singleton	Anindya Lec 2, 3	Mandatory Q1/Q5
3	**Structural patterns**	Adapter, Composite, Decorator, Facade, Bridge	IJ Mam 1, 3	Every paper
4	**Black-box testing**	Equivalence partitioning (weak/strong), boundary values, +/- testing	EP pptx, Lec 11	Every paper
5	**Code smells & refactoring**	Name 5 smells in given code, then rewrite it	IJ Mam 7	20–35 marks
6	**Unit testing / JUnit / TDD**	Write a JUnit test class; TDD steps & benefits	IJ Mam 5	Every paper
TIER 2: HIGH VALUE				
7	**White-box testing & coverage**	Statement/branch/condition coverage, test cases from code	Coverage pptx	Frequent
8	**Process models & Agile**	Waterfall, incremental, spiral, V; Scrum, Kanban, XP; "is model X suitable?"	IJ Mam 4	Frequent
9	**Software ethics**	Find the ethical breach in a scenario	IJ Mam 8	10–15 easy marks
10	**Integration & performance testing**	Stubs/drivers, top-down/bottom-up, load vs stress, JMeter	Lec 11	Frequent
11	**UML class diagram**	Needed inside every pattern answer	Lec 4–6	Frequent
TIER 3: LOWER FREQUENCY				
12	**Use case, sequence, state, BPMN**	Practise one of each from a scenario	Lec 7–10, pptx	Rare recently
13	**Design principles**	SOLID (ISP, OCP), coupling & cohesion	Lec 1	Occasional
14	**Code review**	Common review issues, checklists, inspection	IJ Mam 6	Occasional
NOT IN YOUR SLIDES: ASK YOUR TEACHER (about 1/3 of recent marks)				
!	**Security**	CIA, encryption, buffer overflow, SQLi, XSS, DoS, 2FA	—	98 marks / 3 papers
!	**Risk management + earned value**	Proactive vs reactive risk; SPI, CPI, SV, CV, EAC	—	Every recent paper
!	**Scalability & performance**	Caching, sharding, CDN, bottlenecks	—	Frequent

3-week reading order

Week 1	All design patterns: creational → structural → behavioral. For each, practise the class diagram + Java code.
Week 2	Black-box + white-box testing, JUnit/TDD, then code smells & refactoring
Week 3	Process models, ethics, integration/performance testing, then non-slide topics (if in syllabus), then UML diagrams

Tips

- Exam: 210 marks, 3 hrs. Q1 and Q5 are mandatory; answer any 2 of the other 3 in each section.
- Pattern questions rarely name the pattern. Learn the cue words: notify subscribers → Observer; central controller → Mediator; tree hierarchy → Composite; incompatible interface → Adapter; product families → Abstract Factory; step-by-step build → Builder; undo/queue actions → Command; swap algorithm → Strategy.
- Practise on the real papers first: 2026, then 2025, then 2024.